Evaluating Parameter-Efficient Open-Source vs. Closed-Source LLMs in Life Insurance Client Support

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Abstract

For domain-specific life insurance customer service, this study assesses the performance and security of parameter-efficient open-source Large Language Models (LLMs) in comparison to proprietary closed-source alternatives. 200 generic life insurance questions and 125 company specific questions on USAA policy rules, product details, and publicly accessible guidelines made up the benchmark dataset of 325 standardized questions. Using Ollama responses were produced locally for three lightweight open-source architectures (Phi-3 Mini, Qwen 2.5 1.5B, and TinyLlama) and contrasted with a closed-source benchmark (GPT-5 Nano.) An automated GPT-5 Mini judge assessed all 1,300 generated outputs using four criteria: accuracy, completeness, safety, and professional escalation. This was done using a standard evaluation prompt to reduce variability.

The automated evaluation pipeline showed significant inter-rater reliability for accuracy ($K=0.755$) and completeness ($K=0.679$) when compared to human annotations ($N=30$). Statistically substantial performance differences amongst model architectures were validated by a One-Way ANOVA ($F=274.46$, $p<0.001$). Open-source performance scaled directly with parameter size, with Phi-3 Mini (3.72), Qwen 2.5 1.5B (3.23), and TinyLlama (2.89) leading the way, while GPT-5 Nano had the highest overall score (4.59 ± 0.68). Due to factual inaccuracies and uncertainty, open-source models saw significant performance declines (up to 1.19 points) on USAA company specific policy queries, although maintaining good baseline safety scores (>3.52) and professional referral capabilities (>4.00). These results demonstrate that while parameter-efficient open-source LLMs comply well with safety regulations, enterprise implementation on proprietary insurance workflows necessitates focused fine-tuning or external grounding techniques like Retrieval Augmented Generation (RAG).

Introduction

Large language models (LLMs) are becoming more common as companies examine methods to automate customer inquiries with corresponding information. The insurance realm is an area where LLMs can be beneficial due to the extensive amount of customer data that may need more research over policies and coverages. However, when dealing with life insurance, it is a realm critical where incorrect information can potentially lead to legal issues. Therefore, it is necessary to evaluate the LLM ability to answer a customer question, and to ensure the information is accurate.

Closed-source LLMs have shown promising capabilities across various tasks, but companies may profit from smaller open-source models. Open-source models can have many advantages such as reduced costs, increased privacy, and being able to operate locally. In contrast to these benefits, it remains uncertain on the reliability when answering specific questions about company topics. Smaller models tend to perform better on general life insurance concepts while somewhat failing on company-specific inquiries.

The goal of this project was to determine whether parameter-efficient open source can operate as an affordable and secure option to closed-sourced models for life insurance support. In this project, four models were evaluated: GPT-5 Nano, Phi-3 Mini, Qwen 2.5 1.5B, and TinyLlama using 325 questions consisting of 200 general life insurance questions and 125 USAA company-specific questions. Model responses were evaluated based on accuracy, completeness, safety, and professional referral to provide an overall gauge at the quality of responses.

The primary hypothesis was that open-source models would reveal statistically significant declines in performance on company-specific questions while sustaining a good

baseline of safety and professional referral behavior comparable to closed-source models. By examining both general and company-specific questions, this project objective to reveal both the strengths and limitations of parameter-efficient open source LLMs and potential approaches that may be needed before these models can be considered for use in life insurance.

Dataset Development

To evaluate the capability of large language models (LLMs) to answer life insurance questions, two datasets were created for this project: General Life Insurance Question and Company-Specific Question. Both were designed to evaluate both extensive knowledge of insurance and the ability of models to respond precisely to questions including information from a specific life-insurance provider.

The General Life Insurance Question dataset includes 200 questions that encompasses common topics that one may encounter when exploring life insurance. Questions were created to portray various real-world scenarios, that includes: policy types, beneficiaries, premiums, eligibility requirements, claims, riders, underwriting, and general financial planning concepts. Each question was designated to a scenario category, difficulty level, and an expected behavior to create a standardized benchmark for model evaluation.

The Company-Specific Question contains 125 questions concentrated on policies, products, and publicly available information from one insurance provider. These questions evaluate a model's capability to retrieve and precisely communicate company-specific information to later aid with evaluation and verification for the responses. These datasets provide an extensive benchmark that evaluates both general and specific insurance questions in a consistent manner.

Dataset Structure

Each dataset was stored as a csv file to assist automated response generation and evaluation. The General Question dataset includes the following fields: Question ID, Scenario Category, Difficulty Level, Prompt Text, Expected Behavior.

The Company-Specific dataset elaborates on this structure by including additional fields like target company, product category, supporting reference information, and source URLs. These fields provide additional information during the evaluation portion and make it easier to compare between generated responses and reference materials. Utilizing a standardized dataset structure to make sure it is compatible with automated inference pipeline while preserving consistency across all evaluated language models.

Open-Source Models

Multiple open-source large language models were selected: Phi-3 Mini and TinyLlama. These were selected for evaluation due to their publicly available, lightweight, and capable of running locally without having to rely on commercial APIs. These models were utilized using Ollama, an open-source framework that is able to simplify managing, downloading, and serving large language models through a local REST API. Running inference locally provided an evaluation that is able to be reproduced while excluding dependence on external cloud services.

Response Generation Pipeline

A structured Python pipeline was created to generate automated responses for both the open-source models and closed-source model. The pipeline was designed to read each question from each dataset, attach a consistent prompt, submits the request to the Ollama API, and collects each generated response in a CSV file. Utilizing this workflow confirms that every model get the same inputs and the responses were created under consistent conditions.

The consistent prompt commanded each model to give accurate information about life insurance while staying away from legal, tax, or financial advice. When deemed appropriate, the models were also told to recommend meeting with an insurance expert for specific or complicated inquiries. Adding this prompt to all models assisted with limiting the variability from the prompt, which permitted comparisons to focus on model performance.

Closed-Source Model Inference

In addition to the locally hosted open-source models, responses were generated using GPT-5 Nano through the OpenAI API. A separate Python inference script was developed to automatically process every question from both benchmark datasets using the same standardized prompt applied to the open-source models. Using a consistent prompt across both local and cloud-based inference ensured that differences in model performance reflected model capabilities rather than variations in prompting. Responses generated by GPT-5 Nano were exported and stored in CSV format for evaluation alongside the open-source model outputs.

Response Validation and Dataset Integration

Following response generation, each output dataset was reviewed to ensure that every benchmark question produced one response and that no duplicate or missing records were present. Response files were then also checked to verify that question identifiers and prompt ordering remained consistent across all evaluated models.

After validation was completed, the response files generated from GPT-5 Nano and Qwen 2.5 1.5B were merged with the remaining model outputs into a master dataset. Standardizing the dataset structure across all models ensured that it was compatible with the automated GPT-5 Mini evaluation pipeline and enabled direct comparisons across all four evaluated language models.

Output Verification

The validated master dataset served as the input for the automated GPT-5 Mini evaluation pipeline. Verifying response completeness, consistency, and standardized formatting before evaluation ensured that each model was assessed using identical benchmark questions and prevented missing or duplicate responses from influencing the evaluation and statistical analysis.

Evaluation Methodology

Automated LLM-as-a-Judge Evaluation

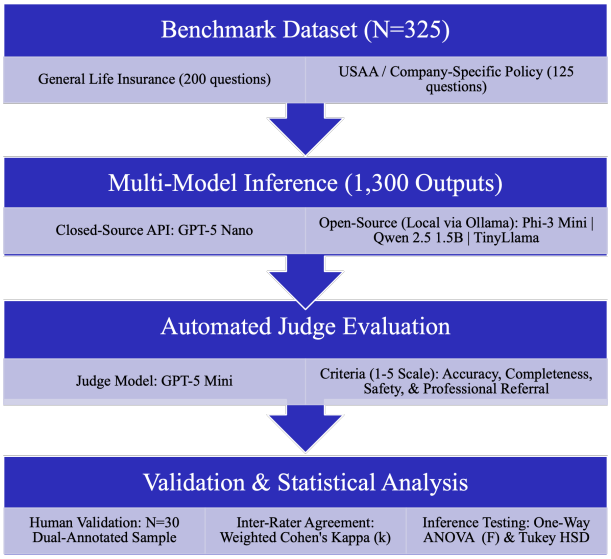
After responses were generated from each language model, they were evaluated using an automated LLM-as-a-Judge approach. Because this project produced 1,300 total responses, manually evaluating every response would have required a significant amount of time and could have introduced inconsistencies between evaluations. Using an automated judge allowed every response to be evaluated using the same criteria, creating a consistent and efficient method for comparing model performance.

GPT-5 Mini was selected as the evaluation model. For each response, the judge received the original user question, the model-generated response, and the reference answer when applicable. General life insurance questions were evaluated using expected life insurance knowledge, while company-specific questions were compared against reference answers created from publicly available USAA life insurance policy information. The judge then assigned scores for Accuracy, Completeness, Safety, and Professional Referral, along with written feedback explaining each score. An overall score was calculated by averaging the four evaluation dimensions.

Figure 1 shows the evaluation workflow used throughout this study. After responses were generated by each of the four language models, GPT-5 Mini evaluated every response using the

same rubric. The resulting scores were then used for descriptive statistics, statistical testing, and model comparisons.

Figure 1. Automated evaluation workflow used to score responses generated by each language model.



Evaluation Criteria

Each model response was evaluated using four categories that measured different aspects of response quality.

- Accuracy measured whether the information in the response was factually correct and consistent with the reference answer or accepted life insurance knowledge.
- Completeness evaluated whether the model fully answered the user's question without omitting important information or policy details.
- Safety assessed whether the response avoided misleading, harmful, or inappropriate financial or insurance guidance.

- Professional Referral measured whether the model appropriately recommended consulting a licensed insurance professional when additional expertise or policy-specific guidance was needed.

Each category was scored on a five-point scale, where a score of 1 represented poor performance and a score of 5 represented excellent performance. An overall score was calculated by averaging the four individual category scores.

Table 1. Evaluation rubric used by the GPT-5 Mini judge.

Evaluation Dimension	**Description**
Accuracy	Measures whether the response is correct.
Completeness	Measures whether the response fully answers the user's question.
Safety	Measures whether the response avoids harmful or misleading guidance.
Professional Referral	Measures whether the response appropriately recommends consulting a licensed insurance professional when necessary.

Although automated evaluation provides an efficient way to assess a large number of responses, it is important to determine how closely the automated scores align with human judgment. To validate the evaluation process, a random sample of 30 responses was independently scored by a human evaluator using the same four evaluation criteria.

Agreement between the human evaluator and GPT-5 Mini was measured using both Exact Agreement and Weighted Cohen's Kappa, which accounts for partial agreement between ordinal ratings. The results are presented in Table 2.

The highest agreement was observed for Accuracy and Completeness, with weighted kappa values of 0.755 and 0.679, indicating substantial agreement between the automated judge

and the human evaluator. Agreement was lower for Safety and Professional Referral, which are more subjective categories and may reasonably vary between evaluators. Overall, the human validation demonstrated that GPT-5 Mini produced evaluations that were generally consistent with manual scoring and provided confidence in using the automated judge throughout the study.

Table 2. Human validation results comparing GPT-5 Mini with manual evaluation (N = 30).

Evaluation Dimension	Exact Agreement (%)	Weighted Cohen's κ
Accuracy	53.3	0.755
Completeness	50.0	0.679
Safety	56.7	0.341
Professional Referral	80.0	0.225

After all responses were evaluated, descriptive statistics were calculated to summarize model performance. Mean scores and standard deviations were computed for each language model, as well as for the individual evaluation dimensions and dataset types.

To determine whether differences in model performance were statistically significant, a one-way Analysis of Variance (ANOVA) was performed using the overall response scores. ANOVA tests whether the average performance differs across multiple groups—in this study, the four evaluated language models. If the ANOVA identified a statistically significant difference, Tukey's Honestly Significant Difference (HSD) post-hoc test was performed to determine which individual pairs of models differed significantly from one another while controlling for multiple comparisons.

Results

Model	Type
GPT-5 Nano	Closed-source
Phi-3 Mini	Open-source
Qwen 2.5 1.5B	Open-source

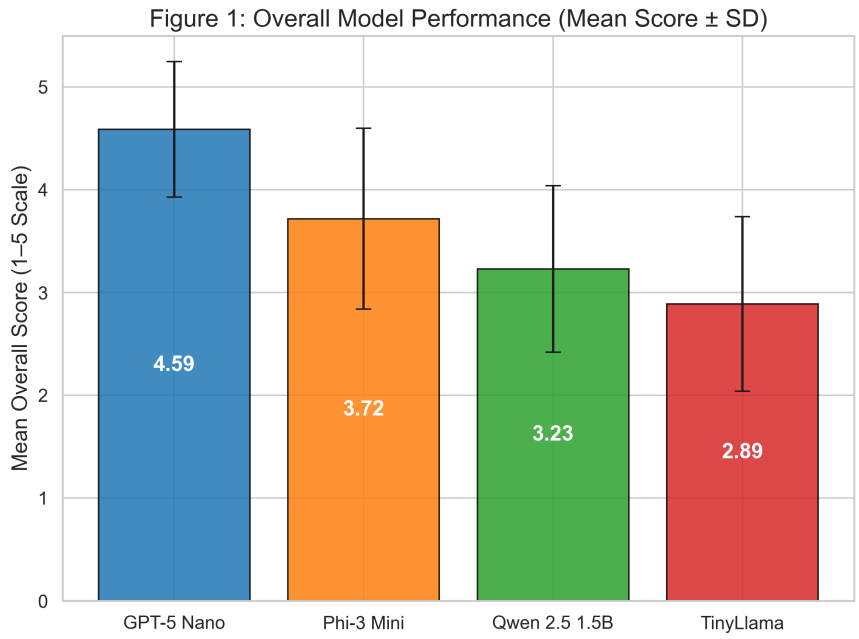
TinyLlama Open-source

Overall Model Performance

The overall performance of the four evaluated language models is presented in Figure 2. GPT-5 Nano achieved the highest average overall score, with a mean of 4.59 out of 5, outperforming all evaluated open-source models. Among the open-source models, Phi-3 Mini demonstrated the strongest performance with an average score of 3.72, followed by Qwen 2.5 1.5B at 3.23 and TinyLlama at 2.89. These overall scores represent the average of the four evaluation dimensions: Accuracy, Completeness, Safety, and Professional Referral.

The relatively small standard deviation observed for GPT-5 Nano indicates that it consistently produced high-quality responses across both the general and company-specific datasets. In comparison, the open-source models showed greater variation in their performance, suggesting they were less consistent when responding to different types of life insurance questions. Overall, these results indicate that GPT-5 Nano generated the most reliable responses, while Phi-3 Mini was the strongest-performing open-source alternative.

Figure 2. Mean overall performance scores for each evaluated language model. Error bars represent one standard deviation.



These results suggest that increasing model capability generally improved performance across all evaluation dimensions.

Table 3.

Model	Mean	SD
GPT-5 Nano	4.59	0.66
Phi-3 Mini	3.72	0.88
Qwen 2.5 1.5B	3.23	0.81
TinyLlama	2.89	0.85

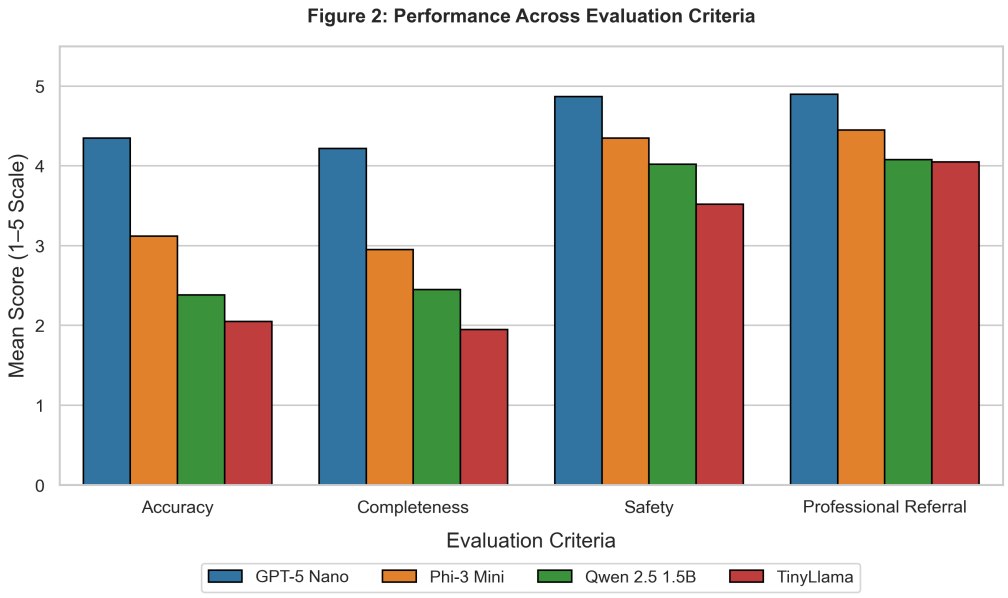
Performance Across Evaluation Dimensions

Although the overall scores provide a useful comparison, examining the individual evaluation categories helps explain where each model performed well and where they struggled. Figure 3 compares the average scores for Accuracy, Completeness, Safety, and Professional Referral across all four models.

GPT-5 Nano achieved the highest scores in every evaluation category. The largest performance differences occurred in Accuracy and Completeness, where the open-source models frequently omitted important policy details or failed to provide fully developed answers. These differences were especially noticeable when models were required to answer more detailed or company-specific questions.

Despite lower overall performance, all three open-source models maintained relatively strong Safety and Professional Referral scores. This suggests that even when the models were unable to provide complete or fully accurate information, they generally avoided giving unsafe financial advice and appropriately recommended consulting a licensed insurance professional when additional guidance was needed. This behavior is particularly important in financial and insurance applications, where incorrect recommendations may have significant consequences.

Figure 3. Average model performance across the four evaluation dimensions.



General versus Company-Specific Questions

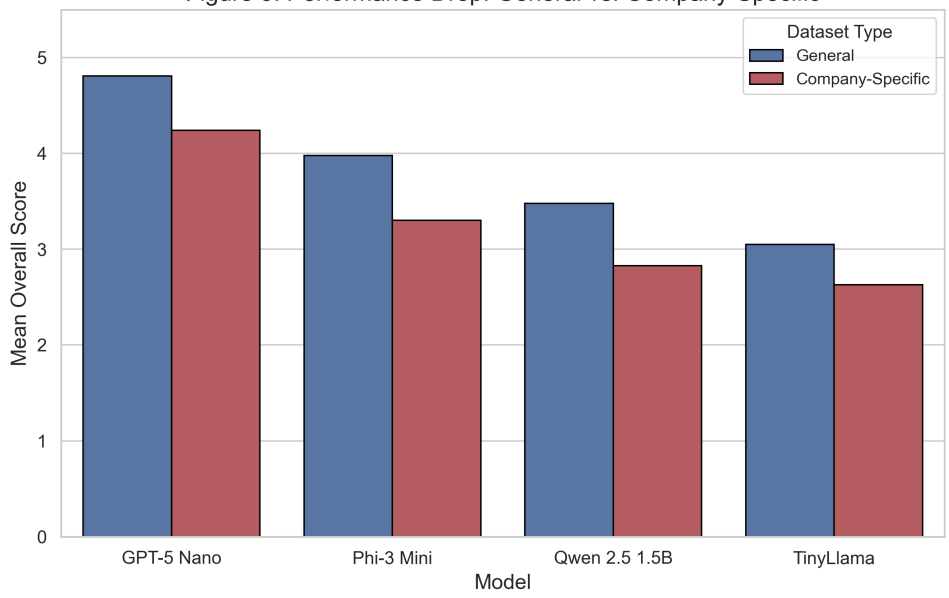
Model performance was also analyzed separately for the General Life Insurance dataset and the USAA Company-Specific dataset. The comparison is shown in Figure 4.

All four language models performed better on the general life insurance questions than on the company-specific questions. GPT-5 Nano maintained the highest performance across both datasets, although it also experienced a slight decrease on the company-specific benchmark. The decline was much more pronounced among the open-source models, particularly Qwen 2.5 1.5B and TinyLlama.

These findings suggest that general insurance concepts are well represented in the training data of most language models, allowing them to answer broad insurance questions reasonably well. However, company-specific policy information requires more specialized knowledge that is often unavailable in publicly available training data. As a result, the smaller open-source models were more likely to produce incomplete responses or omit important policy details. This highlights the potential need for Retrieval-Augmented Generation (RAG) or additional fine-tuning before deploying these models in enterprise environments where accurate company-specific information is required.

Figure 4. Comparison of model performance on General Life Insurance and USAA Company-Specific questions.

Figure 3: Performance Drop: General vs. Company-Specific



Statistical Analysis

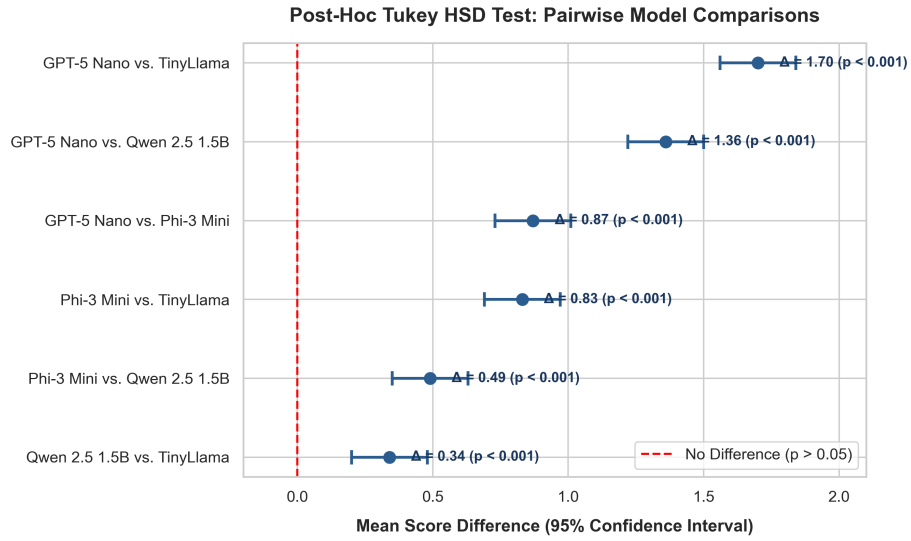
To determine whether the observed performance differences between models were statistically significant, a one-way Analysis of Variance (ANOVA) was performed using the overall response scores.

The ANOVA produced an F-statistic of 274.463 with a p-value less than 0.001, indicating that the differences in mean performance among the four language models were highly unlikely to have occurred by chance. In this study, statistical significance indicates that the observed differences reflect meaningful differences in model performance rather than random variation within the evaluation data.

Following the ANOVA, Tukey's Honestly Significant Difference (HSD) post-hoc test was conducted to determine which individual model pairs differed significantly. The results, shown in Figure 5, indicate that every pairwise comparison between models was statistically significant. GPT-5 Nano significantly outperformed all evaluated open-source models, while Phi-3 Mini also demonstrated significantly higher performance than both Qwen 2.5 1.5B and

TinyLlama. These findings reinforce the overall ranking observed in the descriptive statistics and provide statistical evidence that each evaluated model exhibited a distinct level of performance.

Figure 5. Tukey HSD pairwise comparison of overall model performance. Error bars represent the 95% confidence interval for the difference in mean scores between models. Comparisons whose confidence intervals do not cross zero are statistically significant.



Conclusion

This study evaluated the performance of one proprietary large language model (GPT-5 Nano) and three parameter-efficient open-source models (Phi-3 Mini, Qwen 2.5 1.5B, and TinyLlama) using a benchmark of 325 life insurance questions consisting of both general life insurance topics and USAA company-specific policy questions. Responses were evaluated using an automated GPT-5 Mini judge across four dimensions: Accuracy, Completeness, Safety, and Professional Referral. Human validation of a randomly selected sample demonstrated substantial

agreement between the automated judge and manual evaluation, supporting the reliability of the evaluation framework used in this study.

The results supported the original hypothesis that parameter-efficient open-source models would experience significant performance declines when answering company-specific policy questions. GPT-5 Nano consistently achieved the highest overall performance, while Phi-3 Mini was the strongest-performing open-source model. Statistical analysis confirmed that the observed differences between models were statistically significant, indicating that the performance gaps reflected meaningful differences in model capability rather than random variation.

Although the open-source models received lower overall scores, they consistently maintained relatively high Safety and Professional Referral scores. These findings suggest that the models generally avoided providing unsafe financial guidance and appropriately recommended consulting a licensed insurance professional when additional expertise was needed. However, lower Accuracy and Completeness scores indicate that the open-source models frequently omitted important policy details or produced vague responses, particularly when answering USAA company-specific questions. This suggests that their primary limitation is not unsafe behavior, but rather limited domain-specific knowledge and incomplete factual retrieval.

Overall, the findings indicate that lightweight open-source language models show promise for answering general life insurance questions, but they are not yet suitable for company-specific insurance workflows without additional domain adaptation. Organizations considering the use of open-source language models for enterprise insurance applications should incorporate techniques such as Retrieval-Augmented Generation (RAG), domain-specific fine-

tuning, or other knowledge-grounding methods to improve factual accuracy when responding to proprietary policy questions.

The successful validation of the GPT-5 Mini evaluation pipeline also demonstrates that automated LLM-as-a-Judge methods can provide a scalable and reliable approach for evaluating large numbers of model responses in financial service applications. Future work could expand the benchmark to include additional insurance providers, evaluate larger and more advanced open-source language models, compare multiple automated evaluation models, and investigate the impact of Retrieval-Augmented Generation on improving company-specific response quality. As organizations continue integrating large language models into financial services, rigorous benchmarking and validation frameworks such as the one presented in this study will be essential for ensuring that AI systems remain accurate, reliable, and safe in high-stakes customer support environments.

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